

Phases

AI-Based Phishing and Social Engineering Detection Assistant (PS57)

Companion to [PRD_PhishGuard_PS57.md](#), [ARCHITECTURE_PhishGuard_PS57.md](#), and [RULES_PhishGuard_PS57.md](#). This is the day-by-day execution timeline from team lock through the live demo. Each phase has a deliverable and an exit condition — don't move to the next phase until the exit condition is actually met, not just "mostly done."

Phase 0 — Team Lock & Registration

Window: now → Aug 20, 11:59 PM **Owner:** whole team

- ☐ All 6 members confirmed and roles roughly assigned (Frontend, Backend/LLM, Heuristics, Research/Content, Presentation/Design, PM/Integration).
- ☐ Team registered on the SIH/institute portal, rules compliance checked (e.g. mandatory female representation if applicable).

Exit condition: Registration submitted. This is a hard deadline — nothing else in this file matters if this slips.

Phase 1 — Research & Study

Window: Aug 19 → Aug 21 **Owner:** whole team, self-paced

- ☐ Understand social engineering vs. generic phishing, and why signature/blocklist tools miss it.
- ☐ Understand Levenshtein distance and domain lookalike detection (done).
- ☐ Understand anchor-tag/href mismatch detection.
- ☐ Understand TLD risk scoring.
- ☐ Understand LLM structured-output prompting (getting reliable JSON back).
- ☐ Understand basic PII anonymization concepts (what counts as PII, why it's stripped pre-LLM-call).

- ☐ Look at 1-2 prior SIH-winning PPT structures for tone/depth reference (structure only, not content).

Exit condition: Every team member can explain the core detection approach out loud without reading from a slide.

Phase 2 — Idea PPT (Round 1)

Window: Aug 20 → Aug 22, 11:59 PM **Owner:** Presentation/Design lead + PM, content from whole team

- ☐ Slide 1: Problem Statement & Target Users
- ☐ Slide 2: Proposed Solution & Core Innovation
- ☐ Slide 3: Technical Architecture & Flow
- ☐ Slide 4: Tech Stack, Feasibility & Roadmap
- ☐ Slide 5: Business Impact, Security & Future Scope
- ☐ Each presenter can defend their slide's claims if a judge pushes back (see "judge questions" in Rules §7).

Exit condition: PPT submitted before the deadline. This is the only thing evaluated in Round 1 — no code required.

Phase 3 — Heuristic Engine

Window: Aug 25 (if shortlisted) **Owner:** Heuristics lead (can double with Backend)

- ☐ Levenshtein-based domain lookalike check, against a small protected-brands list.
- ☐ Anchor/href mismatch detector.
- ☐ TLD risk scoring lookup table.
- ☐ Urgency/fear keyword bank + matcher.
- ☐ All four sub-checks combined into a single `heuristicScore` (0–1) per `ARCHITECTURE.md` §2.2.
- ☐ Each sub-check independently testable/callable, per its own file (Rules §3).

Exit condition: Given raw text input, the heuristics module returns a score and signal breakdown — no LLM call needed yet.

Phase 4 — LLM Reasoning Pipeline

Window: Aug 26 **Owner:** Backend/LLM lead

- ☐ PII anonymizer built and wired in **before** any LLM call (non-negotiable, Rules §1).
- ☐ Single Node.js API route (`/api/analyze`) that sends anonymized content to the LLM with the strict JSON system prompt.
- ☐ Response validated as parseable JSON; fallback verdict on parse failure.
- ☐ Low-temperature setting for consistency across repeated runs.
- ☐ Risk Scoring Engine combining heuristic + LLM scores (starting weights 0.4/0.6, per PRD).
- ☐ Failure handling: LLM timeout/error → fall back to heuristic-only score (Architecture §5, Rules §4).

Exit condition: End-to-end pipeline works — raw input in, final combined score + explanation out — even with a bad network connection (fallback path tested, not just happy path).

Phase 5 — UI Dashboard

Window: Aug 27 → Aug 28 **Owner:** Frontend lead(s)

- ☐ Single-page dashboard: paste area, 0–100 gauge, threat-factor badges, explanation panel.
- ☐ 3 sample-case buttons wired to **hardcoded, pre-computed** responses (not live API calls) — mandatory per Rules §1 and §7.
- ☐ Visual polish pass — this is what judges see first; it should look finished, not like a wireframe.

Exit condition: Someone unfamiliar with the project can open the dashboard, click a sample case, and understand the verdict without explanation.

Phase 6 — Demo Testing & Validation

Window: Aug 29 (morning) **Owner:** whole team

- ☐ Run all 3 test cases through the real pipeline (not just the hardcoded fallback) at least once, and confirm results make sense.
- ☐ Case 3 (legitimate message) specifically confirmed as a clean non-flag — treat any false positive here as a blocker (Rules §5).
- ☐ Re-check the 0.4/0.6 weighting against actual test results; adjust if scores look off.
- ☐ Test on venue-like network conditions if possible, not just home Wi-Fi.

Exit condition: All 3 cases produce correct, explainable verdicts, and the sample-case buttons work with no network at all.

Phase 7 — Pitch Practice

Window: Aug 29 (afternoon/evening) **Owner:** whole team, presenter(s) lead

- ☐ Rehearse a crisp 60-second live demonstration.
- ☐ Presenter(s) can answer without notes: "why not just use a blocklist," "what's your false-positive rate," "how do you protect user privacy."
- ☐ Confirm who presents which slide/section and who drives the live demo.
- ☐ Backup plan rehearsed: if live demo has any issue, fall straight to the sample-case buttons.

Exit condition: Full run-through completed at least twice, under time, with no one reading off a script.

Quick Reference Timeline

Date	Phase
Aug 19–20	Phase 0 (team lock) + Phase 1 (research) overlap
Aug 20–22	Phase 2 (PPT)
Aug 25	Phase 3 (heuristics)
Aug 26	Phase 4 (LLM pipeline)
Aug 27–28	Phase 5 (UI)

Date	Phase
Aug 29 AM	Phase 6 (testing)
Aug 29 PM	Phase 7 (pitch practice)